

# Mohammad Rafieian

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## SUMMARY

Software engineer and Ph.D. researcher who builds scalable, secure systems end-to-end. Strong in backend architecture, ML pipelines, and large-scale automation, with deep expertise in program analysis and software security. Experienced shipping Dockerized full-stack applications (Django/React) and designing reproducible, data-intensive infrastructure.

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, JavaScript, Scala, SQL

**Backend & Web:** Django, Flask, Node.js, React, REST APIs, WebSockets

**ML & Data:** PyTorch, TensorFlow, scikit-learn, LLM fine-tuning, ML pipelines

**Infrastructure:** Docker, Linux, Git, PostgreSQL, Redis, Firebase, Vercel

**Focus Areas:** Program Analysis, Automated Testing, Software Security, Backend Systems

## EXPERIENCE

**Graduate Research Assistant** — University of Texas at Dallas

Aug 2023 – Present

- Designed and maintained large-scale, reproducible experimental pipelines using fuzzing and dynamic instrumentation to benchmark four static analysis tools.
- Built ML training and evaluation infrastructure (PyTorch/TensorFlow) with automated hyperparameter tuning and cross-validation, fine-tuning transformer models to +30% accuracy.
- Containerized experiments with Docker on Linux for portability and reproducibility; authored six peer-reviewed papers.

**Research Assistant** — Model-Driven SE Lab, Isfahan University

Jan 2022 – Jan 2023

- Developed an automatic reservation-system generator engine that produced full web applications from high-level models.

## PROJECTS

**ArtCom — Online Art Auction Platform** · React, Django, PostgreSQL, Docker

Feb – Jun 2022

- Built a real-time art auction and exhibition platform (React.js + Django REST Framework), cutting renting costs by 90%.
- Implemented a Dockerized, WebSocket-based bidding system supporting 200 concurrent participants with Redis-backed synchronization.

**LORD — Low-Code Reservation Platform** · Django, React, MySQL

Aug – Dec 2022

- Pioneered a low-code reservation-system generator that automated website creation, cutting deployment costs by over 90%.
- Engineered a scalable full-stack solution with integrated payment processing for seamless deployment.

**Call Graph Pruning Framework** · Python, Java, PyTorch

Feb – Jul 2025

- Fine-tuned CodeBERT/CodeT5 with custom training pipelines to improve call-graph edge classification accuracy by 30%.
- Re-evaluated prior methods and fixed dataset flaws, showing reported gains were inflated (+50% claimed vs. +5% actual).

## EDUCATION

**University of Texas at Dallas**, M.S. & Ph.D. in Software Engineering & Security

2023 – Present

M.S. May 2026 · Ph.D. expected May 2027 · Advisor: Dr. Shiyi Wei

**Isfahan University**, B.S. in Computer Engineering

Aug 2019 – Jun 2023

ACM Student Chapter Member

## RESEARCH & PUBLICATIONS

Author of six papers in top software-engineering venues (ICSE, ICST, ESEM, FSE) on program analysis, automated testing, and ML for security. Full list at mohpydev.github.io.